

MA10207 Analysis 1 - Semester 1, 2021/22

Problem Sheet Week 2 - Solutions

1. Prove that, for all $n \in \mathbb{N}$, $8^{2n} - 6^{2n}$ is a multiple of 7.

Answer. Let $\Lambda = \{n \in \mathbb{N} : 8^{2n} - 6^{2n} \text{ is a multiple of } 7\}$.

Clearly $1 \in \Lambda$ as $8^2 - 6^2 = 28 = 7 \times 4$.

If $n \in \Lambda$, then there exists $m \in \mathbb{N}$ such that $8^{2n} - 6^{2n} = 7m$, and

$$\begin{aligned} 8^{2(n+1)} - 6^{2(n+1)} &= 8^{2n}8^2 - 6^{2n}6^2 \\ &= 8^{2n}8^2 - 6^{2n}8^2 + 6^{2n}8^2 - 6^{2n}6^2 \\ &= (8^{2n} - 6^{2n})8^2 + 6^{2n}(8^2 - 6^2) \\ &= 7 \times m \times 8^2 + 28 \times 6^{2n} = 7(m \times 8^2 + 4 \times 6^{2n}). \end{aligned}$$

Since $(m \times 8^2 + 4 \times 6^{2n}) \in \mathbb{N}$, if $n \in \Lambda$, then $n + 1 \in \Lambda$. By induction, it follows that $\Lambda = \mathbb{N}$. That is the formula is true for all $n \in \mathbb{N}$.

2. Prove that

$$\sum_{k=1}^n \frac{1}{\sqrt{k+1} + \sqrt{k}} = \sqrt{n+1} - 1,$$

for all $n \in \mathbb{N}$.

Answer. Let $\Lambda = \{n \in \mathbb{N} : \sum_{k=1}^n \frac{1}{\sqrt{k+1} + \sqrt{k}} = \sqrt{n+1} - 1\}$.

Clearly $1 \in \Lambda$ as $\frac{1}{\sqrt{2+1}} = \sqrt{2} - 1$.

If $n \in \Lambda$, then

$$\begin{aligned} \sum_{k=1}^{n+1} \frac{1}{\sqrt{k+1} + \sqrt{k}} &= \sum_{k=1}^n \frac{1}{\sqrt{k+1} + \sqrt{k}} + \frac{1}{\sqrt{n+2} + \sqrt{n+1}} \\ &= \sqrt{n+1} - 1 + \frac{1}{\sqrt{n+2} + \sqrt{n+1}} \\ &= \frac{\sqrt{(n+2)(n+1)} + n + 2 - \sqrt{n+2} - \sqrt{n+1}}{\sqrt{n+2} + \sqrt{n+1}} \\ &= \frac{\sqrt{n+2}(\sqrt{(n+1)} + \sqrt{n+2}) - (\sqrt{n+2} + \sqrt{n+1})}{\sqrt{n+2} + \sqrt{n+1}} \\ &= \sqrt{n+2} - 1. \end{aligned}$$

So if $n \in \Lambda$, then $n + 1 \in \Lambda$. By induction, it follows that $\Lambda = \mathbb{N}$. That is the formula is true for all $n \in \mathbb{N}$.

3. Let $A = \{\frac{n-1}{2n} : n \in \mathbb{N}\}$. Prove that A is bounded. Find $\sup(A)$ and $\inf(A)$.

Answer. Since $\frac{n-1}{2n} = \frac{1}{2} - \frac{1}{2n}$ and n is positive, A is bounded above by $\frac{1}{2}$. Furthermore $\frac{n-1}{2n} \geq 0$, hence A is bounded.

We claim that $\sup(A) = \frac{1}{2}$. Suppose $M < \frac{1}{2}$ is an upper bound for A . Now

$$\begin{aligned} \frac{n-1}{2n} \leq M &\Leftrightarrow n-1 \leq 2Mn \\ \Leftrightarrow (1-2M)n &\leq 1 \Leftrightarrow n \leq \frac{1}{1-2M}. \end{aligned}$$

The latter step is valid since $1-2M > 0$. Thus $\mathbb{N}$ is bounded above, which contradicts the Archimedian postulate. Hence, no such M exists. In other words, $\frac{1}{2}$ is the least upper bound for A .

We claim that $\inf(A) = 0$. Indeed, 0 is a lower bound. For any $L > 0$, L cannot be a lower bound for A since for $n = 1$, $\frac{n-1}{2n} = 0$.

4. Using the order axioms in $\mathbb{R}$, prove that

- [a.] For all $a \in \mathbb{R}$, if $a > 0$, then $a^{-1} > 0$, and if $a < 0$, then $a^{-1} < 0$.
[b.] For all $a, b \in \mathbb{R}$, if $a \leq 0$ and if $b \leq 0$, then $ab \geq 0$.

Answer. [a.] Let us consider the case $a > 0$. By contradiction, assume that $a^{-1} < 0$. Axiom **(A.14)** from the Lecture Notes gives that

$$\forall x, y, z \in \mathbb{R}, \quad \text{if } x \leq y \text{ and } z \geq 0 \text{ then } xz \leq yz.$$

Apply axiom **(A.14)** with $x = a^{-1}$, $y = 0$ and $z = a$ to obtain

$$a^{-1}a \leq 0.$$

On the other hand, $a^{-1}a = 1$, hence

$$1 \leq 0$$

which is in contradiction with Proposition 19, part (v), in the Lecture Notes. So this is impossible.

Thus if $a > 0$, then $a^{-1} \geq 0$. But $a^{-1} \neq 0$, because $a \cdot 0 = 0 \neq 1$ by Proposition 18, part (i) and axiom **(A7)**. Hence $a^{-1} > 0$.

If $a < 0$, we use the same arguments to conclude that $a^{-1} < 0$.

[b.] The order axiom **(A12)**, Lectures Notes MA10207 Analysis 1, Semester 1, 2021/22, tells that

$$0 = -a + a \leq -a, \quad 0 = -b + b \leq -b$$

The order axiom **(A14)**, Lectures Notes MA10207 Analysis 1, Semester 1, 21/22, tells that

$$\text{for all } x, y, z \in \mathbb{R}, \quad \text{if } x \leq y, \quad \text{and } z \geq 0, \quad \text{then } xz \leq yz.$$

Choose $x = 0$, $y = -a$ and $z = -b$. We get: for all $a, b \in \mathbb{R}$, if $0 \leq -a$, and $-b \geq 0$, then $0 = 0(-a) \leq (-a)(-b) = ab$. The latter identity follows from Proposition 18, part (v).

Homework.

1. Prove by induction that, for all $n \in \mathbb{N}$, $n^3 + 2n$ is a multiple of 3.

Answer. Let $\Lambda = \{n \in \mathbb{N} : n^3 + 2n \text{ is a multiple of } 3\}$.

Clearly $1 \in \Lambda$ as $1 + 2 = 3$.

If $n \in \Lambda$, then there exists $m \in \mathbb{N}$ such that $n^3 + 2n = 3m$, and

$$\begin{aligned} (n+1)^3 + 2(n+1) &= n^3 + 3n^2 + 3n + 1 + 2n + 2 \\ &= n^3 + 2n + 3(n^2 + n + 1) = 3(m + n^2 + n + 1). \end{aligned}$$

Since $m + n^2 + n + 1$ is a positive integer, we proved that if $n \in \Lambda$, then $n + 1 \in \Lambda$. By induction, it follows that $\Lambda = \mathbb{N}$. That is the formula is true for all $n \in \mathbb{N}$.

2. Using the order axioms in $\mathbb{R}$, prove that
 - a. For all $a, b \in \mathbb{R}$, if $a \leq 0$ and if $b \geq 0$, then $ab \leq 0$.
 - b. For all $a, b, c \in \mathbb{R}$, if $a \leq b$ and if $c \leq 0$, then $ac \geq bc$.

Answer. a. The order axiom **(A14)**, Lectures Notes MA10207 Analysis 1, Semester 1, 21/22, tells that

$$\text{for all } x, y, z \in \mathbb{R}, \quad \text{if } x \leq y, \quad \text{and } z \geq 0, \quad \text{then } xz \leq yz.$$

Choose $x = a$, $y = 0$ and $z = b$. We get: for all $a, b \in \mathbb{R}$, if $a \leq 0$, and $b \geq 0$, then $ab \leq 0 \cdot b = 0$.

b. Since $a \leq b$, then $b - a \geq 0$ by the order axiom **(A13)**. Since $c \leq 0$, then $(b - a)c \leq (b - a) \cdot 0 = 0$ by order axiom **(A14)**. Hence $bc \leq ac$ by order axiom **(A13)**.

3. Let $A = \{\frac{2n+1}{2n} : n \in \mathbb{N}\}$. Prove that A is bounded. Find $\sup(A)$ and $\inf(A)$.

Answer. Since $\frac{2n+1}{2n} = 1 + \frac{1}{2n}$ and n is positive, A is bounded below by 1. In other words, 1 is a lower bound for A . Furthermore, since $\frac{1}{2n} \leq 1$ then $1 + \frac{1}{2n} \leq 2$ for all $n \in \mathbb{N}$. Hence A is bounded.

We claim that $\inf(A) = 1$. Suppose $L > 1$ is a lower bound for A . Now

$$\begin{aligned}\frac{2n+1}{2n} \geq L &\Leftrightarrow 2n+1 \geq 2nL \\ \Leftrightarrow 2n(L-1) \leq 1 &\Leftrightarrow n \leq \frac{1}{2(L-1)}\end{aligned}$$

where the latter inequality is valid since $L > 1$. Thus $\mathbb{N}$ is bounded above, which contradicts the Archimedean postulate. Hence, no such L exists, and 1 is the greatest lower bound for A .

We claim that $\sup(A) = \frac{3}{2}$. Observe first that $\frac{3}{2}$ is an upper bound for A , as $\frac{1}{2n} \leq \frac{1}{2}$ for all $n \in \mathbb{N}$. Henceforth $\frac{2n+1}{2n} = 1 + \frac{1}{2n} \leq 1 + \frac{1}{2} = \frac{3}{2}$. For any $M < \frac{3}{2}$, M cannot be an upper bound for A since for $n = 1$, $\frac{2n+1}{2n} = \frac{3}{2}$. We conclude that $\sup(A) = \frac{3}{2}$.

4. Find a formula for

$$\sum_{n=1}^m \binom{n}{10}$$

for any $m \in \mathbb{N}$, where we intend that $\binom{n}{k}$ equals 0 when $k, n \in \mathbb{N}$ and $n < k$.

Prove by induction the formula you found.

Answer. Recall Pascal's formula

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}, \quad k, n \in \mathbb{N}, \quad 1 \leq k \leq n+1.$$

Hence, taking $k = 11$,

$$\sum_{n=1}^m \binom{n}{10} = \sum_{n=1}^m \left[\binom{n+1}{11} - \binom{n}{11} \right] = \binom{m+1}{11} - \binom{1}{11} = \binom{m+1}{11},$$

since we intend that $\binom{1}{11} = 0$ being $1 < 11$. We found the formula

$$(*) \quad \sum_{n=1}^m \binom{n}{10} = \binom{m+1}{11}, \quad \forall m \in \mathbb{N}.$$

We next prove formula $(*)$ by induction. Let $\Lambda = \{m \in \mathbb{N} : (*) \text{ is valid}\}$.

Clearly $1 \in \Lambda$ as $\sum_{n=1}^1 \binom{n}{10} = 0$ and $\binom{2}{11} = 0$.

If $m \in \Lambda$, then

$$\begin{aligned}\sum_{n=1}^{m+1} \binom{n}{10} &= \sum_{n=1}^m \binom{n}{10} + \binom{m+1}{10} = \binom{m+1}{11} + \binom{m+1}{10} \\ &= \binom{m+2}{11}.\end{aligned}$$

The latter identity follows from Pascal's formula. We proved that if $m \in \Lambda$, then $m+1 \in \Lambda$. By induction, it follows that $\Lambda = \mathbb{N}$. That is formula $(*)$ is true for all $m \in \mathbb{N}$.

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